PH.30102 **FALL 2026**

Quantum Mechanics III

Homework 1

Seonwoo Park Due 2026-09-06 23:59

Topic

Quantum States and Preparation Procedures

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Quantum States and Preparation Procedures. State all assumptions and conventions.
$$\rho \geq 0, Tr\rho = 1$$
2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.
$$p(k|P, M) = Tr(\rho_P E_k)$$
3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.
$$\gamma = Tr(\rho^2)$$

Show enough intermediate work that another student can reproduce the calculation.